PETROCHEMICAL ENGINEERING S.A.
Instrumentation & Control Engineering

ISA-TR20.00.01 / ISA-20

DATA SHEET - ON-OFF VALVE

BALL VALVE

TAG NO.
HS-1200

REV: 1 HOJA: 1 / 1

PROYECTO:	SERVICIO / FUNCIÓN:	FECHA:	ESTADO:
REFINERY EXPANSION PROJECT UNIT 400	**ON-OFF EMERGENCY SHUTDOWN (ESD)**	**30/04/2026**	**APPROVED FOR CONSTRUCTI…**

P&ID REF: **P&ID-402-12-001** | LÍNEA: **4"-HC-4021-A1A-H** | UBICACIÓN: **COMPLEX SECTOR B - HYDROCAR…** | O. COMPRA: **PO-88231-VALV**

GENERAL

100	**TAG NUMBER**		**HS-1200**
101	**Service Description**		**Light Naphtha Feed Cutoff**
102	**Revision / Status**		**Rev 1 / Approved**
103	**P&ID Reference**		**P&ID-402-12-001**
104	**Line Number**		**4"-HC-4021-A1A-H**
105	**Ambient Temperature Range**	°C	**-10 / +45**
106	**Hazardous Area Classification**		**Class I, Zone 1, Gr IIB+H…**
107	**SIL Requirement**		**SIL 2 Certified (IEC 6150…**

CONDITIONS

#	CONDICIÓN DE PROCESO		MÍN	NORM	MÁX
201	**Flow Rate**	kg/h	1200	**3062**	4500
202	**Inlet Pressure (P1)**	bar g	8.5	**10.0**	12.4
203	**Operating Temperature**	°C	15	**42**	65
204	**Design Pressure**	bar g	-	**19.6**	-
205	**Design Temperature**	°C	-	**120**	-
206	**Operating Density**	kg/m³	690	**715**	730
207	**Dynamic Viscosity**	cP	0.45	**0.52**	0.61
208	**Fluid State & Name**		**Liquid / Treated Naphtha**		
209	**Corrosive Components**		**H2S traces < 50 ppm, NACE…**		

LINE SPEC

301	**Pipe Nominal Size**	inch	**4" (DN100)**
302	**Pipe Schedule & Wall**		**SCH 40 / 6.02 mm**
303	**Pipe Material**		**ASTM A106 Gr. B**
304	**End Connection Type**		**Flanged ASME B16.5**
305	**Flange Rating & Facing**		**Class 300 / Raised Face (…**
306	**Flange Finish**	µin	**Ra 125 - 250 AARH**

VALVE BODY

401	**Manufacturer & Model**		**M/S (1) / Trunnion Mounted**
402	**Body Style / Construction**		**2-Piece Split Body, Full …**
403	**Body Material**		**ASTM A216 Gr. WCB (Carbon…**
404	**Ball & Stem Material**		**ASTM A182 F316 / 17-4PH**
405	**Seat Material & Insert**		**Devlon V-API / Primary PE…**
406	**Stem Packing / Seals**		**Graphite Fire-Safe + Vito…**
407	**Fire Safe Certified**		**Yes, API 607 7th Edition …**
408	**Shutoff Leakage Class**		**API 598 Rate A / ISO 5208…**
409	**Face to Face Standard**		**ASME B16.10 Long Pattern**
410	**NACE MR0175 / ISO 15156**		**Compliant**

ACTUATOR

501	**Manufacturer / Model**		**M/S (1) / Pneumatic Heavy…**
502	**Actuator Type**		**Scotch Yoke / Rack & Pini…**
503	**Actuator Action**		**Spring Return (Single Act…**
504	**Fail Position on Loss of Air**		**FAIL CLOSE (FC)**
505	**Air Supply Min / Max**	bar g	**4.5 min / 7.0 max**
506	**Design Sizing Safety Factor**		**1.50 at min air supply**
507	**Max Stroking Time (Close)**	sec	**< 2.5 seconds (ESD)**
508	**Manual Override**		**Declutchable Manual Gear …**
509	**Body / Cylinder Material**		**Hard Anodized Aluminum / …**

SOLENOID

601	**Manufacturer / Model**		**ASCO 551 Series (or equal)**
602	**Configuration & Action**		**3/2 Way, Direct Acting, D…**
603	**Power Supply**		**24 VDC (Nominal, 3.5 W)**
604	**Explosion Proof Rating**		**Ex db IIC T6 Gb / IP66**
605	**Body & Internal Material**		**SS316 Stainless Steel**
606	**Electrical Connection**		**1/2" NPTF Conduit**
607	**Manual Reset / Test**		**Local push-button manual …**

SWITCHES

701	**Switchbox Manufacturer / Model**		**Westlock / TopWorx D-Seri…**
702	**Sensor Element Type**		**Inductive Proximity Senso…**
703	**Switches Quantity**		**2 (1 Open + 1 Closed Indi…**
704	**Visual Position Indicator**		**Beacon High-Visibility 3D…**
705	**Enclosure Protection**		**NEMA 4X / IP67, Ex ia IIC…**
706	**Terminal Strip Points**		**8-Point screw terminals**

ACCESSORIES

801	**Air Filter Regulator (AFR)**		**Included, SS316 with 5µm …**
802	**Quick Exhaust Valve**		**Yes, High-flow 1/2" SS316…**
803	**Tubing & Fittings Material**		**ASTM A269 TP316L, Swagelo…**
804	**Breather / Exhaust Bug Screen**		**Installed on all exhaust …**
805	**Mounting Hardware & Bracket**		**ISO 5211 Standard, 304/31…**
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NOTAS

(900)	M/S: Manufacturer Standard compliant with project specifications SPEC-INST-400-01.
(901)	Valve assembly, actuator and solenoid must be fully integrated and tested in factory per API 598.
(902)	Fire-safe certificate according to API 607 / ISO 10497 must be submitted with documentation.
(903)	Full material test certificates to EN 10204 3.1 mandatory for all wetted and pressure boundary parts.
(904)	Valve tagged with SS316 engraved nameplate fixed with SS316 drive screws.

PREPARED BY	CHECKED BY	APPROVED BY	CLIENT APPROVAL
28/04/2026	29/04/2026	30/04/2026	02/05/2026
J. Rodriguez (Lead Inst. En…	**M. Chen (Senior Process)**	**A. Torres (Project Eng Mgr)**	**H. Smith (Operations Rep)**
EPC Lead	EPC Lead	EPC Lead	PetroChem Inc